

Python Programming Notes (Sample PDF for RAG Demo)

Introduction

Python is a high-level, interpreted programming language known for its simplicity and readability.

Features

Easy syntax, object-oriented programming, dynamic typing, cross-platform support, large standard library, and extensive community support.

Variables

Variables store data. Example: `x = 10`, `name = 'Alice'`.

Data Types

Common data types include `int`, `float`, `str`, `bool`, `list`, `tuple`, `set`, and `dict`.

Operators

Python supports arithmetic, comparison, logical, assignment, and membership operators.

Conditional Statements

Use `if`, `elif`, and `else` to make decisions.

Loops

Use `for` loops to iterate over sequences and `while` loops for repeated execution.

Functions

Functions are reusable blocks of code defined using the `def` keyword.

Object-Oriented Programming

Python supports classes and objects. Key concepts include encapsulation, inheritance, polymorphism, and abstraction.

Lists vs Tuples

Lists are mutable whereas tuples are immutable.

Exception Handling

Use `try`, `except`, `else`, and `finally` to handle runtime errors.

Applications

Python is widely used in web development, data science, AI, machine learning, automation, and DevOps.